Lambda 100 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 18 complete ICs for all stages within ANN10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	**base10k**	0.0321	0.142	1.26	1.01	0.17	0.99	0.0814	1.02	1.21	0	18/18
1 ANN10	**soft10k**	0.658	0.88	1	1	1.3	22.7	1	1.06	84.2	0	18/18
1 ANN10	**long**	0.656	0.911	1	1	0.955	32.3	1	13.3	5.98e+03	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda 100 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 1 complete ICs for all stages within P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
2 P-only DeepONet10	**base10k**	8.74e-03	0.0113	0.165	0.217	0.0395	0.0155	2.22e-03	0.206	0.266	0	18/18
2 P-only DeepONet10	**soft10k**	0.593	0.718	1	1	9.71	28.3	1	1.02	3.14	0	6/18
2 P-only DeepONet10	**long**	0.602	0.66	1	1	7.39	30.2	1	0.997	1.02	0	1/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda 100 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 13 complete ICs for all stages within ANN10 + P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
3 ANN10 + P-only DeepONet10	**base10k**	0.029	0.0496	0.703	0.756	0.166	0.537	0.051	0.988	1	0	18/18
3 ANN10 + P-only DeepONet10	**soft10k**	0.641	1.04	1	1	2.72	26.3	1	1.17	43.5	0	18/18
3 ANN10 + P-only DeepONet10	**long**	0.416	0.919	1	1.44	1.44	19.8	1	8.33	1.52e+03	0	13/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 18, 32, 39, 44, 45, 53, 57, 68, 69, 73, 76, 78. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda 100 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 2 complete ICs for all stages within T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
4 T/P DeepONet10	**base10k**	1.90e-06	1.94e-06	15.3	1.63	2.93e-06	1.26	0.981	0.995	1	0	17/18
4 T/P DeepONet10	**soft10k**	0.179	0.558	8.19e+07	2.02e+08	0.633	3.58e+10	1.24e+05	1	1	0	18/18
4 T/P DeepONet10	**long**	0.125	0.645	4.41e+04	2.22	0.739	4.12e+10	1.69e+05	4.15	1	0	3/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 7, 69. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda 100 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 18 complete ICs for all stages within ANN10 + T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
5 ANN10 + T/P DeepONet10	**base10k**	0.0491	0.0957	0.962	0.82	0.174	0.97	0.066	0.921	0.993	0	18/18
5 ANN10 + T/P DeepONet10	**soft10k**	0.332	0.973	1.42	11.5	2.46	1	0.561	324	2.03e+03	0	18/18
5 ANN10 + T/P DeepONet10	**long**	0.204	0.783	1	3.18	0.925	2.98	0.442	303	1.32e+04	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda 100 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 4 complete ICs for all stages within T/P/ϕ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
6 T/P/ϕ DeepONet10	**base10k**	0.0357	0.0242	44.2	54.8	0.098	200	18.4	4.59	16.3	0	17/18
6 T/P/ϕ DeepONet10	**soft10k**	0.239	0.59	1.04e+08	3.20e+08	0.812	1.17e+10	1.55e+05	1	1	0	16/18
6 T/P/ϕ DeepONet10	**long**	0.121	0.191	3.06	1.24e+03	0.483	2.24e+07	0.986	13.7	5.37e+04	0	4/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 7, 69, 78. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda 100 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 18 complete ICs for all stages within ANN10 + T/P/ϕ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.0633	0.0986	1	0.584	0.195	0.998	0.192	0.998	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.348	0.995	1.78	8.95	2.93	1	0.583	280	2.01e+03	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.669	0.845	2.62	3.78	0.97	18.8	0.999	205	2.12e+04	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.